SANAMREDDY SAI VENKATA REDDY

AI ENGINEER

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PROFESSIONAL SUMMARY

AI Engineer with 3.6 YOE in software engineering, including 2+ years designing and deploying production-level Agentic AI systems, multi-agent workflows, and RAG pipelines using LangChain, LangGraph, and Python (FastAPI) alongside enterprise Java/Spring Boot microservices. Experienced in developing autonomous AI agents with tool-calling capabilities, semantic matching engines, vector search retrieval (Pinecone, FAISS, PGVector), and structured output validation. Recently joined TCS as Systems Engineer on the internal Loan Processing Platform (LPP), completing onboarding and KT and currently delivering on core KYC microservices with AI-assisted verification workflows. Active builder of deployed AI products including LinkedApply Pro (AI automated job application & ATS tailoring Chrome Extension) and DeGhost (real-time computer vision and WebSocket AI cheat detection platform). Strong foundation in Distributed Systems, System Design, and Algorithms (300+ LeetCode problems solved). Certified SAFe 6 Practitioner and Microsoft Azure Fundamentals (AZ-900).

TECHNICAL SKILLS

GenAI & Agentic Frameworks: LangChain, LangGraph, LangSmith, Multi-Agent Orchestration, Autonomous Agents, Tool Calling
Vector Search & Retrieval: RAG Pipelines, Pinecone, FAISS, PGVector, ChromaDB, Semantic Caching, Hybrid Search (Dense + BM25)
LLMs & Foundation Models: Claude 3.5 Sonnet, GPT-4o, Gemini 1.5 Pro, Llama 3, Structured Outputs (Pydantic), Hugging Face
Backend & APIs: Python (FastAPI, Flask, asyncio), Java (Spring Boot, Spring Integration), RESTful APIs, WebSockets, Microservices
Databases & Caching: PostgreSQL, asyncpg, Redis (Semantic & In-Memory Caching), MongoDB, MySQL, SQLAlchemy
Cloud & DevOps: AWS (Lambda, S3, EC2, IAM), Microsoft Azure (AZ-900), Docker, Jenkins, CI/CD, Git
Testing & Monitoring: LangSmith (Tracing & Evaluation), JUnit, Mockito, PyTest, Splunk Centralized Logging, SonarQube
Tools & Environments: Git, GitHub Copilot, Postman, Swagger, IntelliJ IDEA, VS Code, Chrome Extension (MV3), Figma
CS Concepts: Distributed Systems, System Design, Data Structures and Algorithms, OOP, Design Patterns, SOLID Principles

WORK EXPERIENCE

Tata Consultancy Services (TCS) — Systems Engineer | Internal Project — August 2026 – Present | Hyderabad, India

Project: Loan Processing Platform (LPP)

- Recently joined TCS as Systems Engineer, completing enterprise onboarding and Knowledge Transfer (KT) for the Loan Processing Platform (LPP), an internal microservice framework built on Spring Boot and Spring Integration, exploring integration of AI-assisted automation pipelines.
- Currently working on KYC (Know Your Customer) service tasks, implementing core business logic, data modeling, and database operations for automated customer onboarding, verification, and AI-powered document intelligence checks.
- Engaged in standardizing microservices development through reusable components across security, logging, request/response handling, caching, circuit breakers, and external integrations (BRE, CIBIL, DMS), evaluating LLM-driven semantic verification workflows.

Mphasis Ltd — Associate Software Engineer | Client: FedEx — Apr 2023 – July 2026 | Hyderabad, India

Projects: Enterprise Modernization Automation & Global Regulatory Compliance (GRCS)

- Architected an Agentic AI automation workflow using LangGraph to assist in legacy code modernization, automatically analyzing legacy business logic and generating structured Domain-Specific Language (DSL) scripts to produce corresponding Spring Boot microservices.
- Built an AI-Based Test Case Generation Engine leveraging LLMs and FAISS vector search over historical JIRA tickets, automating test generation and cutting manual QA authoring effort by ~40%.
- Developed a RAG-powered compliance scanning engine with semantic intercept rules (MHC/SLC) to validate international package manifests against customs regulations across 70,000+ daily package scans.
- Engineered high-throughput RESTful APIs using Python FastAPI and Spring Boot for Filter, Split, and Super Split microservices across 120+ global HUB sorting facilities.
- Leveraged Splunk and LangSmith for centralized tracing, telemetry, correlation IDs, and token evaluation, reducing average operational incident resolution time from ~2 hours to ~1 hour.

INTERNSHIP

Murven Design Solutions — Python Intern Nov 2022 – Jan 2023 | Remote, India

- Developed RESTful APIs using Python Flask for file upload, download, and AI-based background removal features for a design platform.
- Integrated AWS cloud services including Lambda, S3, EC2, and IAM for serverless processing and scalable storage.
- Deployed Stable Diffusion ML models via Google Colab for AI-powered image generation workflows.

PERSONAL PROJECTS (0-TO-1 AI PRODUCTS)

LINKEDAPPLY PRO — Agentic AI Job Auto-Applier & Real-Time Resume Tailorer | Chrome Web Store May 2026 – Present

Tech: Python, FastAPI, TypeScript, React, Chrome Extension (MV3), Multi-LLM (Claude/GPT-4o/Gemini), BYOK AI

- Engineered an autonomous AI browser extension that parses job descriptions in real-time, extracts core technical requirements, and dynamically tailors resume content to maximize ATS semantic keyword matching.
- Implemented multi-step form automation using client-side BYOK (Bring Your Own Key) architecture, eliminating server infrastructure GPU cost while guaranteeing zero user data retention.
- Integrated dynamic DOM observers and automated field mapping across major job boards (LinkedIn, Greenhouse, Lever), achieving 95%+ auto-fill accuracy.

DEGHOST — Real-Time AI Interview Integrity & Cheat Detection Platform | Live Demo April 2026 – Present

Tech: Python, FastAPI, WebSockets, OpenCV, PostgreSQL (asyncpg), SQLAlchemy, HTMX, Alpine.js, FPDF2

- Architected a real-time computer vision and heuristics detection engine identifying 65+ stealth AI assistance overlays, teleprompters, and virtual video feeds during live technical interviews (<150ms latency).
- Built an asynchronous FastAPI & WebSocket backend featuring live candidate integrity scoring (0–100%), automated threshold risk flagging, and instant tamper-evident PDF forensic audit report generation.
- Deployed scalable containerized MVP on Render with PostgreSQL (asyncpg) and SQLAlchemy, supporting concurrent real-time audio/video assessment sessions.

TECHNICAL ACHIEVEMENTS & CERTIFICATIONS

- Certified SAFe 6 Practitioner & Microsoft Certified: Azure Fundamentals (AZ-900).
- Published LinkedApply Pro to Google Chrome Web Store, automating job applications with client-side BYOK AI at zero recurring infrastructure cost.
- Engineered and deployed DeGhost live MVP on Render, architecting real-time WebSocket candidate integrity scoring against 65+ AI cheating tools.
- Solved 300+ problems on LeetCode using Java and Python, with expertise across algorithms, dynamic programming, and system design.
- Authored research paper on Lung Cancer Prediction Model using deep learning (ResNet-50) for medical imaging, published in IJARBEST and awarded Best Paper Award.

EDUCATION

BACHELOR OF TECHNOLOGY — Information Technology 2018 – 2022 | CGPA: 7.97

Pondicherry University

Coursework: Artificial Intelligence, Machine Learning, Data Structures and Algorithms, Database Systems, Operating Systems, Computer Networks